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Solution Sheet 1: Logical Operators

Task 1

(5 Points)

Disproving Associativity

The operator $\star$ is a NAND gate: it outputs 1 unless both inputs are 1. To check associativity, we compute the truth tables for $(A \star B) \star C$ and $A \star (B \star C)$.

A	B	C	$A \star B$	LHS: $(A \star B) \star C$	$B \star C$	RHS: $A \star (B \star C)$
0	0	0	1	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	1	1
0	1	1	1	0	0	1
1	0	0	1	1	1	0
1	0	1	1	0	1	0
1	1	0	0	1	1	0
1	1	1	0	1	0	1

The columns for LHS and RHS are not identical. For example, in the second row ($A = 0, B = 0, C = 1$), LHS is 0 while RHS is 1. Thus, the operator is **not associative**.

Task 2

(5 Points)

Proving $A \star A \equiv \neg A$ Compute the truth table for $A \star A$ and compare to $\neg A$.

A	$\neg A$	$A \star A$
0	1	1
1	0	0

The columns match exactly. Intuitively, $A \star A$ is 1 when A is 0 (since NAND of 0 and 0 is 1), and 0 when A is 1 (NAND of 1 and 1 is 0), which is exactly $\neg A$.

Task 3

(10 Points)

Solution for Task 3

One possible formula is:

$$G = A \star (B \star C)$$

Derivation

Instead of guessing, we can systematically derive it. First, express F using standard operators, then translate to $\star$.

Analyzing F 's truth table:

- When $A = 0$, $F = 1$ always. This suggests a structure like $(\neg A) \vee \dots$
- When $A = 1$, F matches $B \wedge C$.

So, $F \equiv (\neg A) \vee (B \wedge C)$.

Verification:

- If $A = 0$, $(\neg 0) \vee \dots = 1 \vee \dots = 1$, correct.
- If $A = 1$, $(\neg 1) \vee (B \wedge C) = 0 \vee (B \wedge C) = B \wedge C$, correct.

Now, translate to $\star$ using equivalences:

- $\neg X \equiv X \star X$
- $X \wedge Y \equiv \neg(X \star Y) \equiv (X \star Y) \star (X \star Y)$
- $X \vee Y \equiv \neg X \star \neg Y$

Substituting:

$$\begin{aligned} (\neg A) \vee (B \wedge C) &\equiv (A \star A) \vee \neg(B \star C) \\ &\equiv \neg(A \star A) \star \neg\neg(B \star C) \\ &\equiv \neg\neg A \star (B \star C) \\ &\equiv A \star (B \star C) \end{aligned}$$

Thus, $G = A \star (B \star C)$ works.

Task 4

(8 Points)

Truth Table for $(A \wedge B) \rightarrow (\neg C \vee A)$

A	B	C	$A \wedge B$	$\neg C$	$\neg C \vee A$	$(A \wedge B) \rightarrow (\neg C \vee A)$
0	0	0	0	1	1	1
0	0	1	0	0	0	1
0	1	0	0	1	1	1
0	1	1	0	0	0	1
1	0	0	0	1	1	1
1	0	1	0	0	1	1
1	1	0	1	1	1	1
1	1	1	1	0	1	1

Explanation:

- $A \wedge B$: True only when both A and B are true
- $\neg C$: Negation of C
- $\neg C \vee A$: True when either C is false OR A is true (or both)
- $(A \wedge B) \rightarrow (\neg C \vee A)$: An implication is false only when the antecedent is true and the consequent is false. Since $(A \wedge B)$ is true only in the last two rows, and $\neg C \vee A$ is true in both those cases, the implication is always true (it's a tautology).

Task 5

(7 Points)

Proving De Morgan's Law: $\neg(A \wedge B) \equiv \neg A \vee \neg B$

A	B	$A \wedge B$	$\neg(A \wedge B)$	$\neg A \vee \neg B$	Match?
0	0	0	1	1	✓
0	1	0	1	1	✓
1	0	0	1	1	✓
1	1	1	0	0	✓

The truth values match in every row, so the equivalence holds.

Intuitive Explanation:

- $\neg(A \wedge B)$ means “it’s not the case that both A and B are true”
- $\neg A \vee \neg B$ means “either A is false, or B is false (or both)”
- These express the same logical condition: the conjunction fails exactly when at least one of the operands is false.